Bayesian Optimization as Design of Experiments

The Entropy Connection

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Where this comes from

- I teach Experimental Design in the KU Leuven MSc Statistics and Data Science.
- Bayesian optimization was not in it, and students kept asking.
- Bolting on a lecture called "and now, BO" was not what I wanted.
  It had to be integrated into the course material.

Some insights from that lecture.

Three choices behind a design

	Classical DOE	**Bayesian optimization**
Model	low-order polynomial	Gaussian process
Target	coefficients, prediction D-, I-optimality	the optimum $x^\star$ expected improvement, entropy search
Rule	design, then augment	one run at a time, adaptively

Both columns are usually taught as one method.

Nothing forces the three rows to travel together.

Targeting the optimum with a polynomial

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \varepsilon, \qquad x^\star = -\frac{\beta_1}{2\beta_2}$$

- Nobody fits this because they want β_1.
- Delta method: $\mathrm{Var}(\hat{x}^\star) \approx \sigma^2\, c^\top (\mathbf{X}^\top \mathbf{X})^{-1} c, \quad c = \partial x^\star / \partial \beta.$
- Minimising it is **c-optimality**, and it predates all of this.

No Gaussian process anywhere.

Information about $x^\star$, whatever the model

We want runs that tell us about $x^\star$. The model stays a separate choice.

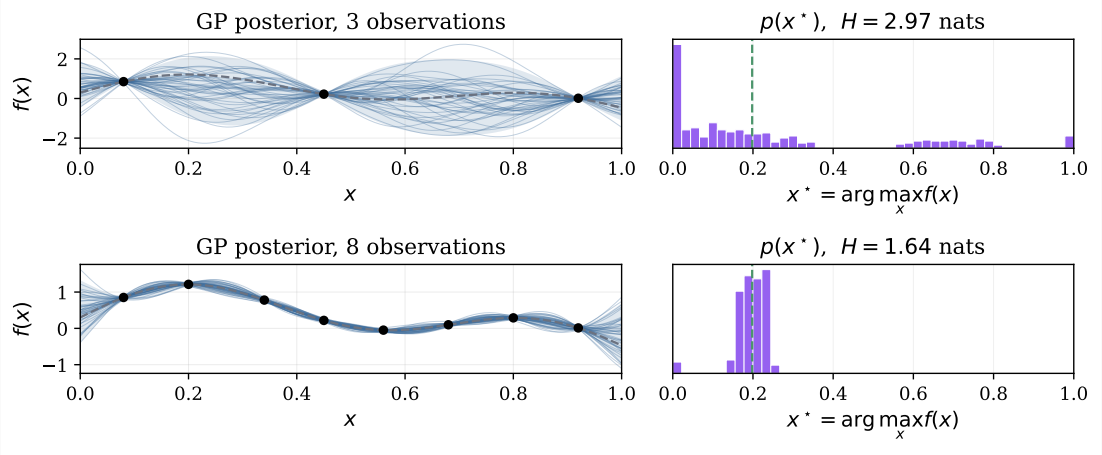
For anyone who was in this room a few hours ago, this is Bayesian experimental design:

$$U(d) \;=\; H\big[p(\psi)\big] - \mathbb{E}_y\Big[H\big[p(\psi \mid y, d)\big]\Big] \;=\; I(\psi;\, y \mid d), \qquad \psi = g(\theta).$$

- $\psi = x^\star$ is Bayesian optimization.
- Familiar criteria fit too. $\psi = \beta$, with a flat prior, is D-optimality. Lindley (1956).

Only ψ changes.

$x^\star$ has a posterior



So it has an entropy, and we can design to reduce it.

Why not just use expected improvement?

Do. It is a good default, particularly for standard BO.

Entropy Search, PES and Max-value Entropy Search score $I(x^\star; y)$ instead. They have not been shown to be clear winners, and they cost far more to compute.

For now their value is as a teaching tool. They say what the acquisition functions are approximating.

I think there is more to be had here.

Hennig & Schuler (2012); Wang & Jegelka (2017).

A sidebar on adaptivity

Standard BO picks the next run one step ahead.

BO with an entropy criterion is a form of Bayesian experimental design, so adaptivity comes for free.

That was my talk this morning.

Foster et al. (2021).

Are Gaussian processes good priors?

BO is Bayesian experimental design with $\psi = x^\star$. The model stays a free choice.

The default is a stationary Gaussian process. Roughly, what it says:

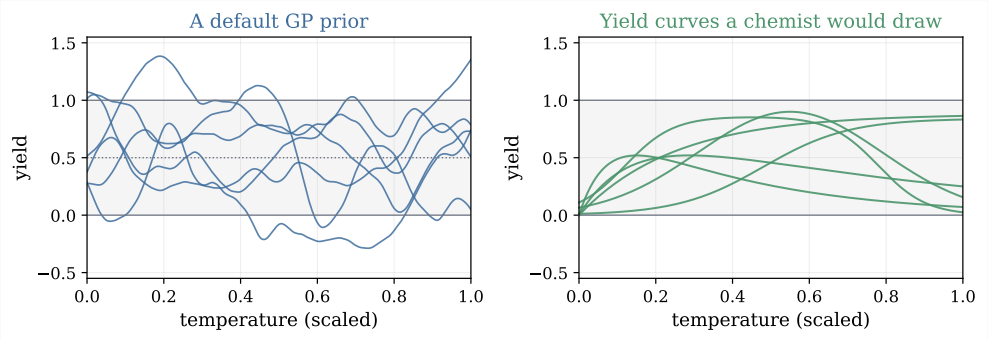
- nearby points are similar,
- everything is pulled back toward a baseline,
- random shocks of the same size everywhere.

Iron concentration in the ground behaves like that. Kriging came from ore bodies.

A random walk with a pull to the mean is a good prior for a rock.

Krige (1951); Matheron (1963).

Is it a good prior for a factory?



A yield is bounded, smooth, single-peaked, and shaped like the last ten runs.

In my opinion, few processes in a factory look like the left-hand panel.

The second-order model is a very strong prior

It allows

- one stationary point, so no plateau and no two settings that compete,
- the same curvature everywhere in the window,
- growth without bound away from the centre.

With a flat prior on β, nothing shrinks it back.

No wonder practitioners reach for tools that model nonlinear behaviour.

Put the second-order model in the mean function

A Gaussian process has a mean function and a kernel. The defaults leave the mean constant and give the kernel all of the work.

Universal kriging. Write the mean as the model you already trust and let the kernel take the residual:

$$f(x) = \beta_0 + \beta^\top x + x^\top B x + \eta(x), \qquad \eta \sim \mathcal{GP}(0, k).$$

- The kernel only explains what the polynomial missed, so ℓ is easier to identify.
- A mechanistic model can take the polynomial's place, leaving the kernel to model what the mechanism gets wrong.

$x^\star$ is now the arg max of the sum, so its posterior carries the uncertainty in β and in η at once.

$I(x^\star; y)$ weighs both, by how much each actually moves $x^\star$.

Experimenting with experimenting

We tell students to use classical DOE when the model form is known, and Bayesian optimization with a Gaussian process when it is not. That split is an accident of 1998.

Both sit in the same space of choices:

- a model, which you should be able to defend as a prior,
- a target, $\psi = g(\theta)$,
- a rule for spending the budget.

This talk recombined them: a polynomial aimed at the optimum, an arg max with a posterior, a Gaussian process with a second-order mean.

Stop seeing BO and DOE as fixed recipes. Start experimenting with experimenting.

Jones, Schonlau & Welch (1998).

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